

Backpropagation: Understanding the implications of the chain rule

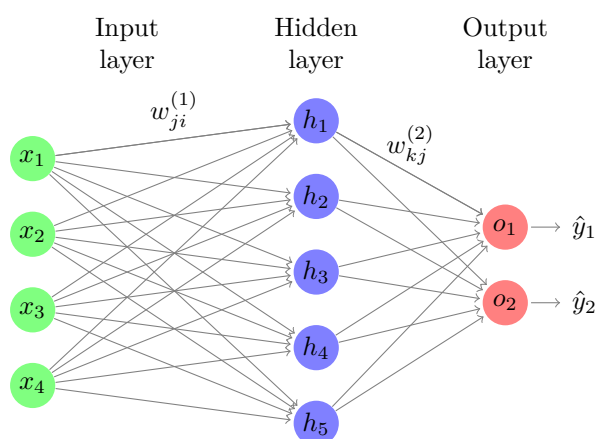
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A lot of the ideas in this lecture come from Andrej Karpathy's blog post on backprop (<https://medium.com/@karpathy/yes-you-should-understand-backprop-e2f06eab496b>) and his CS231n Lecture Notes (<http://cs231n.github.io/optimization-2/>)

Topics

- A quick look at an MLP again
- The chain rule (again)
- Unintuitive gradient effects
- A closer look at basic stochastic gradient descent algorithms

The bias-term free Multilayer Perceptron



Without loss of generality, we can write the above as:

$$\hat{\mathbf{y}} = g(f(\mathbf{x}; \mathbf{W}^{(1)}); \mathbf{W}^{(2)}) = g(\mathbf{W}^{(2)} f(\mathbf{W}^{(1)} \mathbf{x}))$$

where f and g are activation functions.

Gradients of our simple bias-term free MLP

- Let's assume MSE Loss

$$\ell_{MSE}(\hat{\mathbf{y}}, \mathbf{y}) = \|\hat{\mathbf{y}} - \mathbf{y}\|_2^2$$

- What are the gradients?

$$\nabla_{\mathbf{W}^*} \ell_{MSE}(g(\mathbf{W}^{(2)} f(\mathbf{W}^{(1)} \mathbf{x})), \mathbf{y})$$

- Clearly we need to apply the chain rule (vector form) multiple times
- We could do this by hand
- We're not that crazy! We've already seen how AD reduces this to a problem of computing JVPs (or better VJPs here - the reverse mode)
- But, what does that mean in terms of an algorithm for computing the gradients?

Let's go back to a simpler expression

$$\begin{aligned}f(x, y, z) &= (x + y)z \\ &\equiv qz \text{ where } q = (x + y)\end{aligned}$$

Clearly the partial derivatives of the subexpressions are trivial:

$$\begin{aligned}\partial f / \partial z &= q & \partial f / \partial q &= z \\ \partial q / \partial x &= 1 & \partial q / \partial y &= 1\end{aligned}$$

and the chain rule tells us how to combine these:

$$\begin{aligned}\partial f / \partial x &= \partial f / \partial q \cdot \partial q / \partial x = z \\ \partial f / \partial y &= \partial f / \partial q \cdot \partial q / \partial y = z\end{aligned}$$

so $\nabla_{[x,y,z]} f = [z, z, q]$

A computational graph perspective

$$f(x, y, z) = (x + y)z$$

An intuition of the chain rule and reverse mode AD

- Notice how every operation in the computational graph given its inputs can immediately compute two things:
 1. its output value
 2. the *local* gradient of its inputs with respect to its output value
- The chain rule tells us literally that each operation should take its local gradients and multiply them by the gradient that *flows* backwards into it

This is backpropagation

- The backprop algorithm is just the idea that you can perform the forward pass (computing and caching the local gradients as you go),
- and then perform a backward pass to compute the total gradient by applying the chain rule and re-utilising the cached local gradients
- Backprop is just another name for 'Reverse Mode Automatic Differentiation'...

Unintuitive effects I: Multiplication

- Consider the multiplication operation $f(a, b) = a \times b$.
- The gradients are clearly $\partial f / \partial b = a$ and $\partial f / \partial a = b$.
 - (in a computational graph these would be the local gradients w.r.t the inputs)
- If a is large and b is tiny the gradient assigned to b will be large, and the gradient to a small.
- This has implications for e.g. linear classifiers ($\mathbf{w}^\top \mathbf{x}_i$) where you perform many multiplications
 - the magnitude of the gradient is directly proportional to the magnitude of the data
 - multiply $\mathbf{x}_i$ by 1000, and the gradients also increase by 1000
 - if you don't lower the learning rate to compensate your model might not learn
 - **Hence you need to always pay attention to data normalisation!**

Unintuitive effects II: vanishing gradients of the sigmoid

- It used to be popular to use sigmoids (or tanh) in the hidden layers...
- Gradient of $\sigma(x) = \sigma(x)(1 - \sigma(x))$
- Thus as part of a larger network where this is the local gradient, if x is large (+ve or -ve), then all gradients backwards from this point will be zero due to multiplication of the chain rule
 - Why might x be large?
- Maximum gradient is achieved when $x = 0$ ($\sigma(x) = 0.5$, $dx = 0.25$)
 - This means that the maximum gradient that can flow out of a sigmoid will be a quarter of the input gradient
 - * What's the implication of this in a deep network with sigmoid activations?

Unintuitive effects III: dying ReLUs

- Modern networks tend to use ReLUs
- Gradient is 1 for $x > 0$ and 0 otherwise
- Consider $\text{ReLU}(\mathbf{w}^\top \mathbf{x})$
 - What happens if $\mathbf{w}$ is initialised badly?
 - What happens if $\mathbf{w}$ receives an update that means that $\mathbf{w}^\top \mathbf{x} < 0 \forall \mathbf{x}$?
- These are dead ReLUs - ones that never fire for all training data
 - Sometimes you can find that you have a large fraction of these
 - if you get them from the beginning, check weight initialisation and data normalisation
 - if they're appearing during training, maybe λ is too big?

Unintuitive effects IV: Exploding gradients in recurrent networks

- Recurrent networks apply a function recursively for some number of timesteps
- Often this recursion involves a multiplication at each timestep, the gradients of which are all multiplied together because of the chain rule...
- Consider $z = a \prod_n^\infty b$
 - $z \rightarrow 0$ if $|b| < 1$
 - $z \rightarrow \infty$ if $|b| > 1$
- Same thing happens in the backward pass of an RNN (although with matrices rather than scalars, so the reasoning applies to the largest eigenvalue)